

Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Research Paper

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Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Research Paper

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Submitted by:

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Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

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- Python
- Pandas
- Numpy
- Matplotlib
- Scikit-learn
- Streamlit

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I would like to express my sincere gratitude to Unified Mentor for providing me with the opportunity to complete this Data Analytics internship project.

This project helped me gain practical knowledge in data analytics, machine learning, and dashboard development. Working on this project allowed me to improve my technical skills and apply theoretical concepts to a real-world business problem.

I am also thankful to everyone who supported and encouraged me during the completion of this project.

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Abstract

The real estate industry involves a wide variety of buyers with different purchasing behaviors, investment goals, and financial preferences. Understanding these differences is essential for developing effective marketing strategies and providing suitable property recommendations. This project focuses on applying machine learning techniques to segment buyers based on their demographic and investment characteristics.

The project uses data preprocessing techniques such as data cleaning, feature engineering, label encoding, and feature scaling to prepare the dataset for analysis. The K-Means clustering algorithm is used to identify distinct buyer groups, while the Elbow Method and Silhouette Score are used to determine and evaluate the optimal number of clusters.

An interactive dashboard was developed using Streamlit to visualize buyer segmentation, geographic distribution, acquisition purposes, referral channels, loan applications, and other important business insights. The dashboard enables users to filter and explore the data dynamically, supporting informed business decisions.

The findings demonstrate that machine learning can effectively identify hidden customer segments, allowing real estate companies to improve customer targeting, optimize marketing strategies, and better understand buyer behavior. This project highlights the practical application of data analytics and machine learning in solving real-world business problems.

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1. Introduction

The real estate industry generates a large amount of customer and property-related data every day. Understanding buyer behavior is essential for real estate companies to improve customer satisfaction, increase sales, and develop effective marketing strategies. Buyers differ in terms of their investment goals, financial capacity, geographical preferences, and purchasing behavior. Identifying these differences manually is difficult due to the large volume of data.

Machine learning provides an effective solution for analysing customer data and discovering hidden patterns. Clustering techniques group similar customers based on their characteristics without requiring predefined labels. These customer segments help businesses understand buyer preferences and make data-driven decisions.

In this project, the K-Means clustering algorithm is used to segment buyers into different groups based on demographic and investment-related features. Data preprocessing techniques such as data cleaning, feature engineering, label encoding, and feature scaling were applied before building the clustering model. The results were visualized through an interactive Streamlit dashboard, enabling users to explore buyer segments and gain valuable business insights.

1.1 Problem Statement

Real estate companies often treat all buyers as a single customer group despite significant differences in purchasing behaviour, investment objectives, and financial preferences. This lack of customer segmentation results in inefficient marketing campaigns, generic property recommendations, and missed business opportunities.

The objective of this project is to develop a machine learning-based buyer segmentation system that identifies distinct customer groups using demographic and investment-related data. The generated buyer segments can support more effective marketing strategies, improve customer targeting, and assist in making data-driven business decisions.

1.2 Objectives

The main objectives of this project are:

- To preprocess and clean the real estate buyer dataset.
- To perform feature engineering for better model performance.
- To apply label encoding and feature scaling techniques.
- To identify the optimal number of buyer clusters using the Elbow Method.
- To build a buyer segmentation model using the K-Means clustering algorithm.
- To evaluate clustering quality using the Silhouette Score.
- To develop an interactive Streamlit dashboard for visualizing buyer segments and business insights.
- To support data-driven decision-making through machine learning and data visualization.

1.3 Scope of the Project

This project focuses on analysing buyer data from the real estate sector using machine learning techniques. The scope includes data preprocessing, feature engineering, customer segmentation, and visualization of business insights through an interactive dashboard.

The developed system enables users to analyse buyer characteristics, acquisition purposes, geographic distribution, loan applications, and customer satisfaction. The project demonstrates how clustering algorithms can help real estate companies understand customer behaviour, improve marketing strategies, and provide more personalized property recommendations.

2. Dataset Description

2.1 Dataset Overview

The dataset used in this project contains information about real estate buyers and property transactions. It includes demographic details, buyer preferences, property information, financial attributes, and customer satisfaction scores. The dataset was provided as part of the Data Analytics Internship project and was used to perform buyer segmentation using machine learning techniques.

The dataset contains **7,305 records** and **21 features**, providing sufficient information to analyse buyer behaviour and identify meaningful customer segments.

2.2 Dataset Features

The dataset consists of the following features:

Feature	Description
client_id	Unique identifier for each client
client_type	Type of buyer (Individual or Corporate)
first_name	Buyer's first name
last_name	Buyer's last name
date_of_birth	Buyer's date of birth
gender	Gender of the buyer
country	Country of residence
region	Geographic region of the buyer
acquisition_purpose	Purpose of property purchase (Investment or Personal Use)
satisfaction_score	Customer satisfaction rating
loan_applied	Indicates whether the buyer applied for a loan
referral_channel	Source through which the buyer was referred
listing_id	Unique property listing identifier
tower_number	Tower or building number
transaction_date	Date of property transaction
unit_category	Category of the property unit
unit_number	Property unit number

floor_area_sqft	Floor area of the property (in square feet)
sale_price	Final sale price of the property
listing_status	Current status of the property listing
client_ref	Client reference code

2.3 Dataset Summary

The dataset contains both categorical and numerical features. Categorical variables include client type, country, region, acquisition purpose, referral channel, and loan status. Numerical variables include satisfaction score, floor area, sale price, and buyer age.

Before applying the machine learning model, the dataset was cleaned by removing unnecessary columns, handling data types, creating an Age feature from the date of birth, converting the sale price into numeric format, and preparing the data through feature encoding and scaling. These preprocessing steps ensured that the dataset was suitable for clustering analysis.

3. Methodology

The methodology followed in this project consists of a systematic process for preparing the dataset, building the machine learning model, evaluating the clustering performance, and developing an interactive dashboard. The overall workflow is illustrated through the following stages:

- Data Collection
- Data Cleaning
- Feature Engineering
- Label Encoding
- Feature Scaling
- K-Means Clustering
- Cluster Evaluation
- Dashboard Development

3.1 Data Collection

The dataset used in this project was provided as part of the Data Analytics Internship. It contains buyer information, property details, transaction records, and customer-related attributes. The dataset consists of 7,305 records and 21 features representing demographic, financial, and real estate transaction information.

The data was imported into Python using the Pandas library for further preprocessing and analysis.

3.2 Data Cleaning

Data cleaning was performed to improve the quality and reliability of the dataset before applying machine learning algorithms.

The following preprocessing steps were carried out:

- Checked the dataset structure and data types.
- Verified that there were no missing values.
- Confirmed that no duplicate records were present.
- Converted the **sale_price** column from currency format into a numeric data type.
- Converted the **date_of_birth** column into datetime format.
- Calculated the buyer's **Age** from the date of birth.

- Removed unnecessary columns such as client names, client reference numbers, listing identifiers, and transaction dates since they were not required for clustering.

These steps ensured that the dataset was clean and suitable for further analysis.

3.3 Feature Engineering

Feature engineering was performed to improve the quality of the data before applying the machine learning model. A new feature named **Age** was created by calculating the difference between the current year and the buyer's date of birth. This feature provides meaningful information about the buyer's age, which plays an important role in customer segmentation.

Additionally, the **sale_price** column was converted into a numerical format by removing the currency symbol and commas, allowing it to be used for analysis and clustering.

3.4 Label Encoding

Since machine learning algorithms cannot process categorical text values directly, label encoding was applied to convert categorical variables into numerical values.

The following columns were encoded:

- Client Type
- Gender
- Country
- Region
- Acquisition Purpose
- Loan Applied
- Referral Channel
- Unit Category
- Listing Status

Label encoding transformed each unique category into a corresponding numerical value while preserving the information required for clustering.

3.5 Feature Scaling

The dataset contains features with different value ranges. For example, sale price and floor area have much larger values than satisfaction score. To ensure that all features contribute equally during clustering, feature scaling was performed using the **StandardScaler** technique.

StandardScaler transforms the data so that each feature has a mean of zero and a standard deviation of one. This prevents features with larger values from dominating the clustering process.

3.6 K-Means Clustering

K-Means Clustering was selected as the machine learning algorithm for buyer segmentation. It is an unsupervised learning algorithm that groups similar data points into clusters based on their characteristics.

The algorithm assigns each buyer to the nearest cluster centroid and repeatedly updates the cluster centres until the optimal grouping is achieved. In this project, K-Means was used to identify buyers with similar purchasing behaviour, investment preferences, and demographic characteristics.

The resulting clusters help real estate companies understand different buyer segments and support data-driven marketing strategies.

3.7 Elbow Method

The Elbow Method was used to determine the optimal number of clusters for the K-Means algorithm.

In this method, the Within-Cluster Sum of Squares (WCSS) is calculated for different numbers of clusters. The point where the decrease in WCSS begins to slow down forms an "elbow," indicating the most suitable number of clusters.

Based on the Elbow Method, the optimal number of clusters selected for this project was **four (4)**.

3.8 Silhouette Score

The Silhouette Score was used to evaluate the quality of the generated clusters.

The Silhouette Score measures how similar a data point is to its own cluster compared to other clusters. The score ranges from **-1 to 1**, where values closer to **1** indicate better-defined clusters.

For this project, the obtained Silhouette Score was approximately **0.12**, indicating moderate cluster separation. Although the score is not very high, it provides meaningful customer segments that can support business decision-making and buyer profiling.

4. Results and Discussion

The machine learning model successfully segmented real estate buyers into different clusters based on demographic, financial, and investment-related characteristics. The clustering results were analysed using various visualizations and an interactive Streamlit dashboard. These visualizations provide meaningful insights into buyer behaviour and investment patterns.

4.1 Elbow Method

The Elbow Method was used to determine the optimal number of clusters for the K-Means clustering algorithm. The Within-Cluster Sum of Squares (WCSS) values were calculated for different numbers of clusters. The resulting graph showed a noticeable bend at $K = 4$, indicating that four clusters provide the best balance between clustering accuracy and model simplicity.

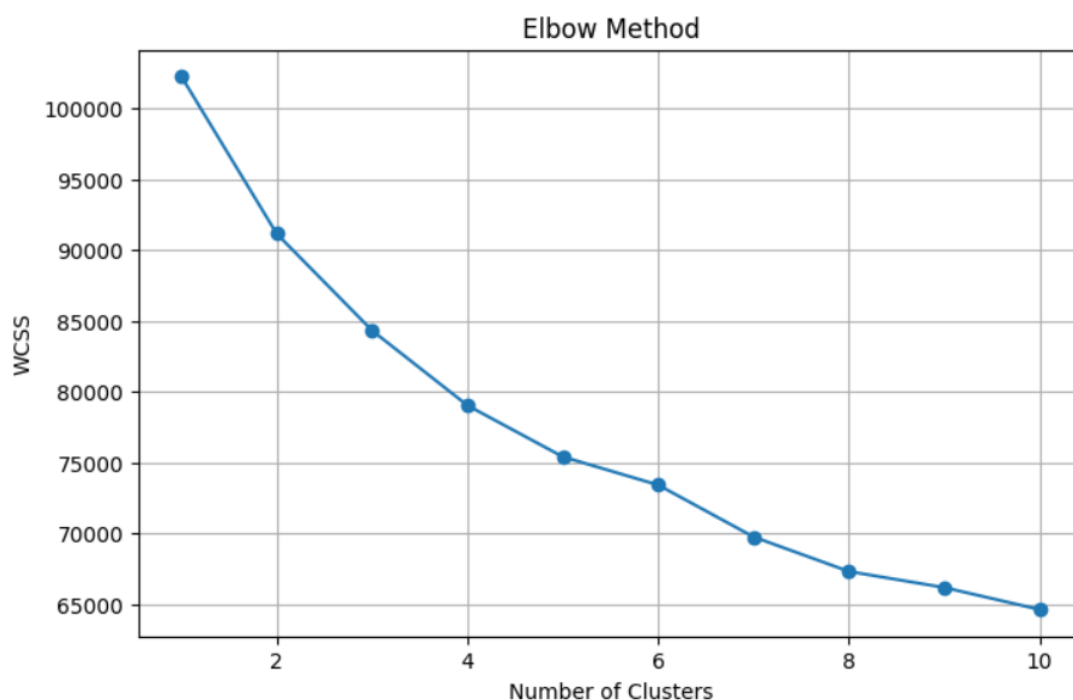


Figure 4.1: Elbow Method Graph

4.2 Buyer Cluster Distribution

The buyer cluster distribution shows the number of buyers assigned to each cluster by the K-Means algorithm. The distribution indicates that some buyer groups contain significantly more customers than others, reflecting differences in buyer behaviour and purchasing patterns.

This visualization helps identify the largest customer segments and supports business planning for targeted marketing strategies.

Buyer Cluster Distribution

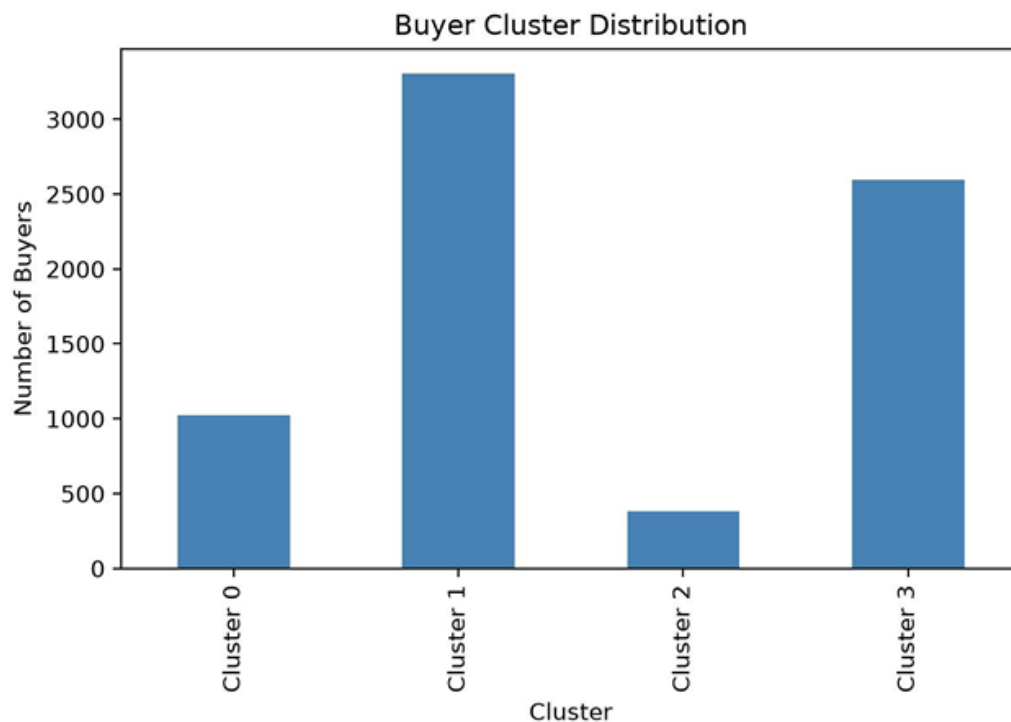


Figure 4.2: Buyer Cluster Distribution

4.3 Buyer Type Distribution

The Buyer Type Distribution compares the number of Individual and Corporate buyers in the dataset. The analysis provides an overview of the composition of the customer base and helps understand the proportion of different buyer categories.

This information can assist real estate companies in designing marketing campaigns specifically for individual customers or corporate investors.

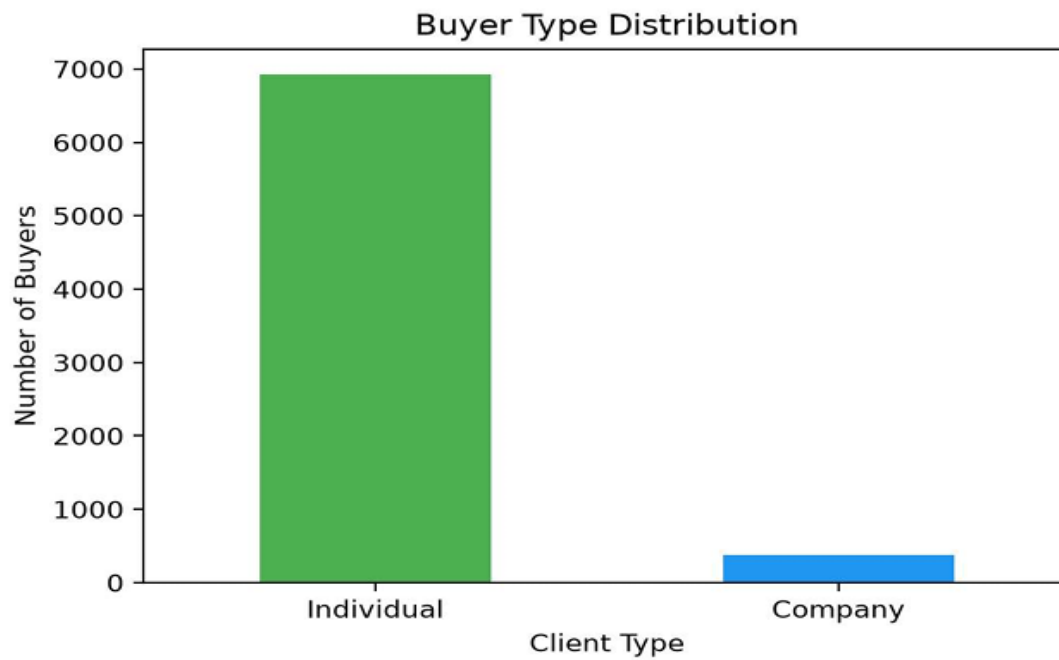


Figure 4.3: Buyer Type Distribution

4.4 Acquisition Purpose Analysis

This analysis illustrates the primary reasons for purchasing properties, such as investment or personal use. Understanding acquisition purposes helps businesses identify customer intentions and recommend suitable property options.

The results indicate the dominant purchase objective among buyers in the dataset.

Acquisition Purpose Analysis

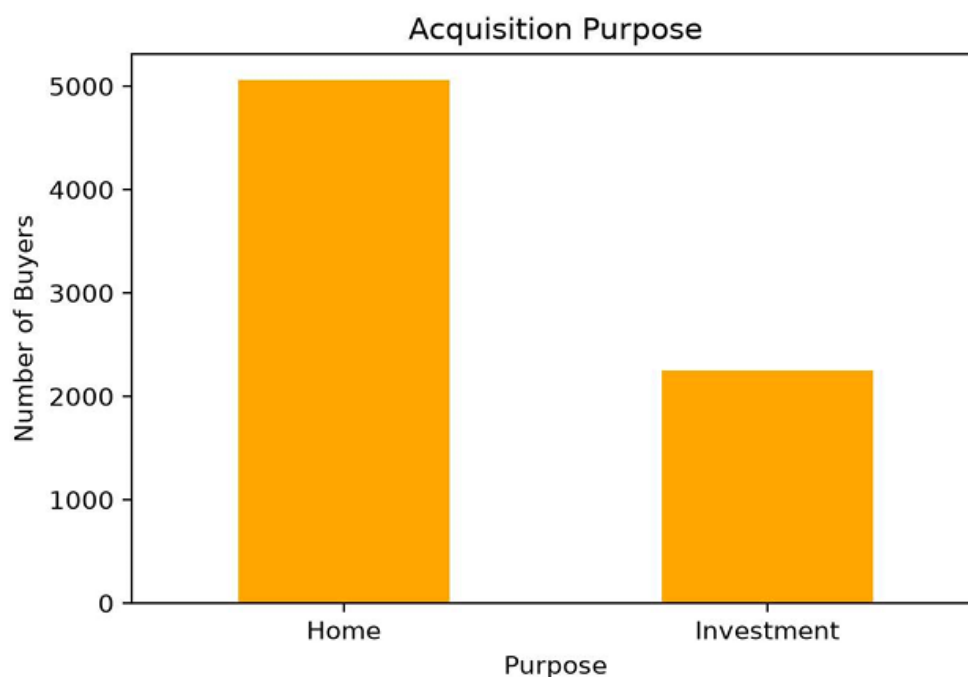


Figure 4.4: Acquisition Purpose Analysis

4.5 Average Sale Price by Cluster

The Average Sale Price by Cluster analysis compares the average property prices across the identified buyer segments. This visualization highlights the purchasing capacity of different customer groups.

The analysis shows that some clusters are associated with higher-value property purchases, indicating premium buyers and investors, while other clusters represent buyers with relatively lower purchasing budgets.



Figure 4.5: Average Sale Price by Cluster

4.6 Geographic Buyer Analysis

The Geographic Buyer Analysis presents the distribution of buyers across different regions. This visualization helps identify regions with higher customer activity and stronger real estate demand.

Understanding regional buyer distribution enables companies to focus marketing efforts, allocate resources effectively, and identify high-potential markets for future investments.

Geographic Buyer Analysis

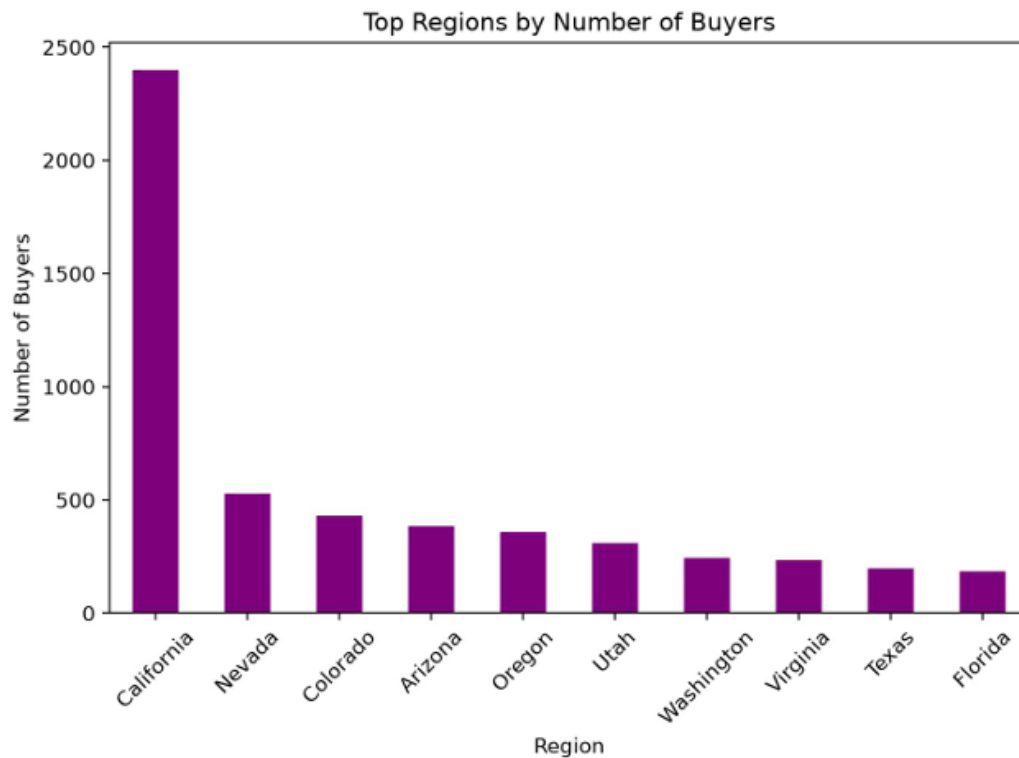


Figure 4.6: Geographic Buyer Analysis

4.7 Loan Applied Analysis

The Loan Applied Analysis compares buyers who applied for financial assistance with those who purchased properties without loans.

This analysis provides valuable information about customer financing behaviour and helps real estate companies understand the dependence of buyers on home loans and financing options.

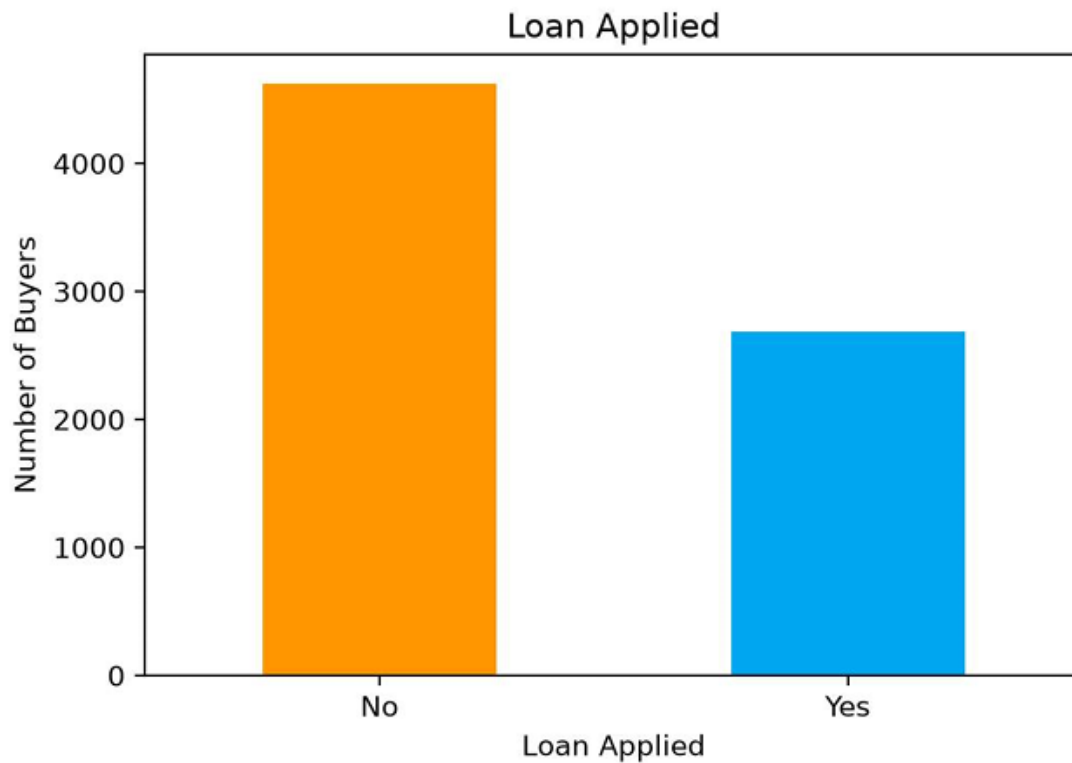


Figure 4.7: Loan Applied Analysis

4.8 Referral Channel Analysis

The Referral Channel Analysis identifies the primary sources through which buyers were introduced to the real estate company. Common referral channels include websites, advertisements, agents, and customer referrals.

The results help evaluate the effectiveness of different marketing channels and support future marketing investment decisions.

Referral Channel Analysis

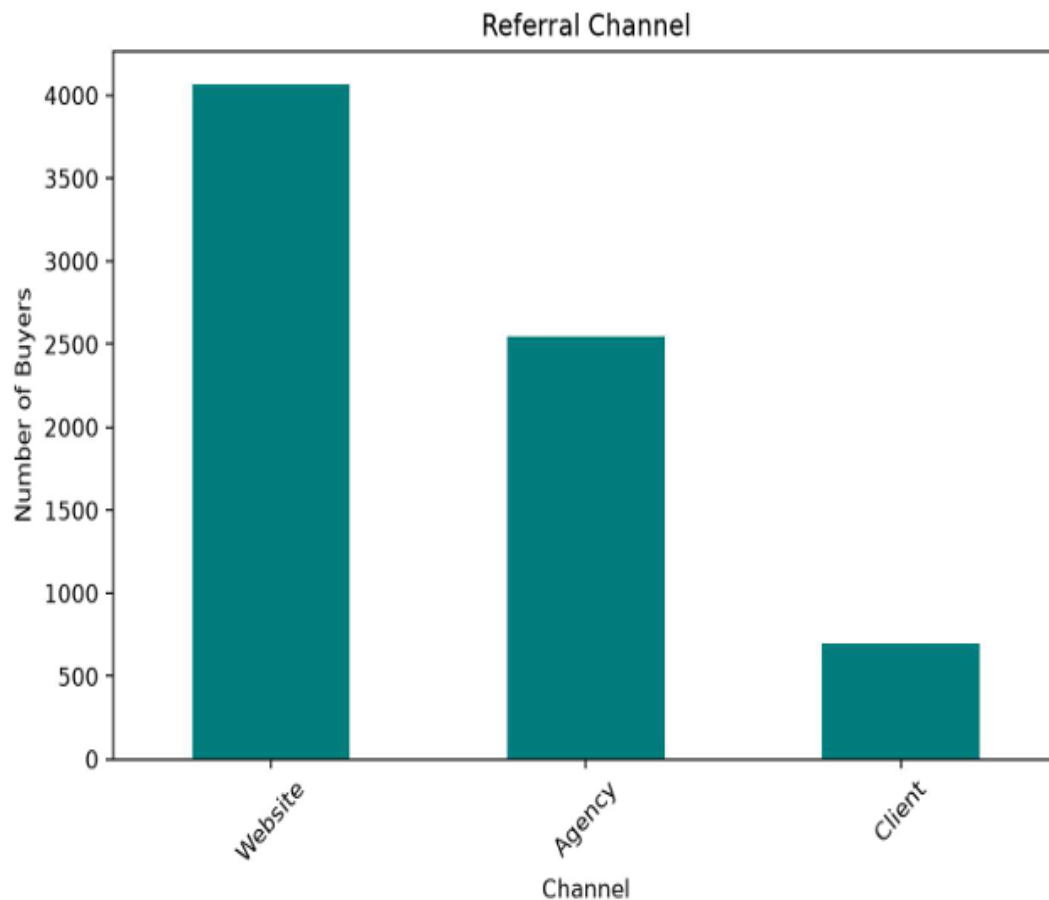


Figure 4.8: Referral Channel Analysis

4.9 Average Satisfaction Score by Cluster

Customer satisfaction was analysed across the identified buyer clusters. Comparing satisfaction scores helps determine which customer segments experience higher satisfaction with the services provided.

This analysis supports improvements in customer experience by identifying buyer groups that may require additional attention or personalized services.

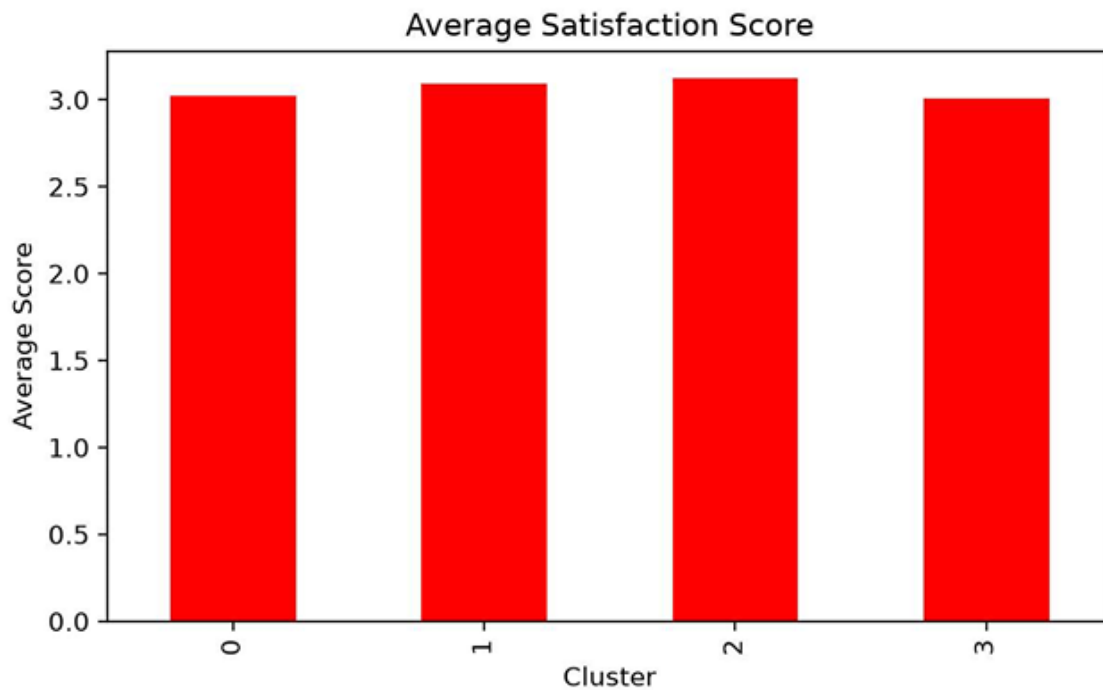


Figure 4.9: Average Satisfaction Score by Cluster

4.10 Business Insights

Based on the analysis and machine learning results, the following business insights were obtained:

- The K-Means algorithm successfully segmented buyers into four distinct customer groups.
- Certain buyer clusters were associated with higher average property prices, indicating premium investment behaviour.
- Buyer acquisition purposes were primarily divided between investment and personal use, helping identify customer motivations.
- Geographic analysis revealed regions with greater buyer concentration, supporting region-specific marketing strategies.
- Loan application analysis highlighted the financing preferences of different customer groups.
- Referral channel analysis identified the most effective customer acquisition sources.
- Satisfaction analysis provided insights into customer experience across different buyer segments.

These findings demonstrate how machine learning and data analytics can support better customer understanding, targeted marketing, and informed business decision-making in the real estate sector.

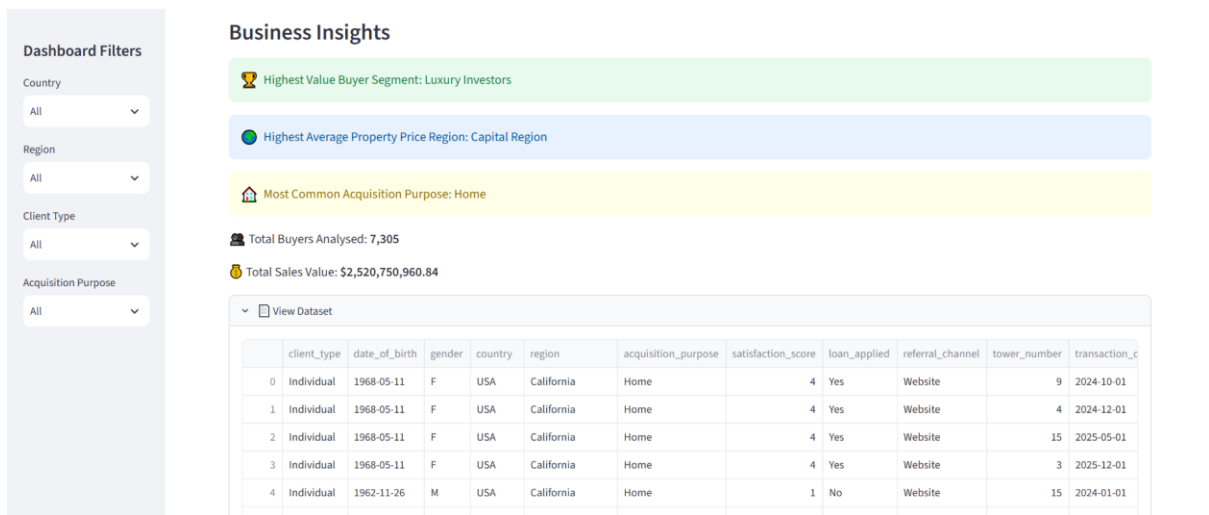


Figure 4.10: Business Insights

5. Conclusion

This project successfully demonstrated the application of machine learning techniques for buyer segmentation in the real estate sector. By performing data cleaning, feature engineering, label encoding, and feature scaling, the dataset was prepared for clustering analysis.

The K-Means clustering algorithm effectively grouped buyers into four distinct segments based on their demographic and investment characteristics. The Elbow Method was used to determine the optimal number of clusters, while the Silhouette Score was used to evaluate the clustering performance.

An interactive Streamlit dashboard was developed to visualize buyer segments, property sales, geographic distribution, loan applications, referral channels, and customer satisfaction. The dashboard enables users to analyse buyer behaviour dynamically and supports data-driven business decisions.

Overall, this project demonstrates how machine learning and data analytics can help real estate companies better understand customer behaviour, improve marketing strategies, and provide more personalized services.

5.1 Future Scope

The project can be further enhanced by incorporating additional buyer and property attributes such as income level, occupation, investment history, and property preferences to improve segmentation accuracy.

Advanced machine learning algorithms such as DBSCAN, Hierarchical Clustering, or Gaussian Mixture Models can also be explored and compared with K-Means clustering. Future work may include predictive models for buyer behaviour, property recommendation systems, and real-time dashboard integration using cloud databases.

The Streamlit dashboard can also be expanded with interactive maps, advanced visualizations, and live data updates to provide deeper business insights for real estate companies.

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